

Chun-Yu (Echo) Shih

Product Engineer · React · TypeScript · FastAPI · Product Ownership

Taipei, Taiwan (UTC+8) · Remote — APAC hours or async

echoshihtw@gmail.com · linkedin.com/in/echoshihtw · github.com/echoshihtw · echoshihtw.github.io/portfolio

Summary

Product-minded full-stack engineer who turns ambiguous operational needs into shipped software — React/Next.js frontends, FastAPI services, data models, testing, and CI/CD. Uses controlled AI-assisted workflows, keeping architecture and verification explicit.

Experience

Lockerbie Technology (Musaeum Ltd, UK group)

Full-Stack Engineer — Internal Platform · Taipei, Taiwan · Dec 2024 – Present

- Built and operate a project-management platform for the **UK construction sector**, taking client organisations and vendor contractors through a project-to-tender-to-quotation lifecycle governed by **ISO 19650** principles. One of two engineers and sole author of the **Next.js / TypeScript** frontend; ~5 users, 10 projects.
- Implemented the team's **event-driven modular monolith** — bounded modules over **RabbitMQ**, project relationships in **ArangoDB** — both new to me, both now in production.
- Given a requirement that layers stay independent, chose **Next.js App Router** and introduced a **BFF trust boundary**: browser clients call same-origin API routes only, backend tokens stay in **HttpOnly cookies**, and **CSRF** is enforced centrally on unsafe requests.
- Designed the six-role permission model — org-level **RBAC** plus project-scoped **PBAC** — resolving roles into permissions and capability flags so services never reimplement access logic. Covered with **Vitest**, **React Testing Library**, and **pytest**.
- Wrote the platform's product specification: role model, architecture rules, eight milestones across three phases, exit criteria, and a decision log — and held scope with a feature freeze until Phase 1 shipped.
- Brought a teammate with no software background into the codebase by pairing — 278 co-authored commits — from first contribution to shipping independently within two months.
- Built a local **Claude Code** workflow — reusable skills and automated hooks — to speed implementation and testing, with final verification left to deterministic quality checks.
- Owned the path from laptop to production in both repos: one-command **Makefile** setup, staging/production deploy bundles, multi-stage **Docker** builds with cached layers, and build-once, promote-by-digest releases (**GitHub Actions**, **GHCR**) cutting three environment builds to one.

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Full Stack Engineer & Founding Partner · Singapore · Remote · Aug 2024 – Oct 2024

- Resolved a macOS **Electron code-signing, notarisation, and auto-update** release blocker, then documented the deployment workflow for repeatable releases across a distributed team.

Find Recruiter

Software Engineer · Taipei, Taiwan · Aug 2022 – May 2024

- Led frontend delivery for a **B2B recruitment and ATS platform** in **React and TypeScript**, in a two-engineer product team reporting to the CTO.
- Drove an incremental **JavaScript-to-TypeScript** migration and built **100+ reusable components**, improving consistency and reducing type-related regressions across a large production codebase.

Independent / Freelance Engineering

Product Engineer · Remote · 2020 – Present

- Built products across fintech, research, e-commerce, and creative industries, including the **Next.js frontend for a Bitcoin coin-selection research tool** with a **BDK core maintainer**.

Projects

Clio — *Founder / Product Engineer* · Bilingual **React/Supabase** salon operations product in daily use at a Tainan salon **since May 2025**, supporting 603 clients and 2,694 service records with **Postgres RLS** and per-member permissions.

Runway — *Founder / Product Engineer* · Personal financial-runway app in **Flutter: Clean Architecture** across a Melos monorepo, **SQLCipher**-encrypted local storage, 7-language localisation. In development — running on Android, iOS distribution in progress.

Skills

Frontend — React · Next.js · TypeScript · TanStack Query · Zustand · Redux · MUI · Tailwind CSS · Flutter · Dart

Backend & Data — FastAPI · Python · Node.js · Postgres · ArangoDB · Supabase · RabbitMQ · SQLite

Infra & Quality — Docker · GitHub Actions · GHCR · AWS · Vitest · React Testing Library · pytest · Claude Code

Desktop — Tauri · Electron

Education

B.A. Italian Language and Culture — Fu Jen Catholic University, Taipei · 2019